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Modelling the Effects of Climate Change on Community Mental Health Care Service Use in Australia

Background

Climate change is a global issue influencing weather patterns around the world. As of 2017, global warming has reached around 0.8°C - 1.2°C above pre-industrial levels and continues to increase each year due to human activities (Masson-Delmotte, 2022). These higher temperatures exacerbate the occurrence of acute natural disasters and have widespread consequences on agriculture (e.g. from climate and precipitation variability), biological ecosystems and the spread of injury and disease (Rawat et al., 2023).

Of noted importance is the increased burden on mental health (MH) due to the pervasive nature of climate change across multiple social and environmental domains (World Health Organization, 2024). Higher ambient temperatures (or surface air temperatures) are associated with an overall increase in population mental health burden of disease (Liu et al., 2025), aggression and violent suicides (Lin et al., 2008). For example, in the United States, higher temperatures are found to intensify existing mental illnesses and disorders, with an increased risk among males, those aged 40+ years, uninsured peoples, Native and African Americans (Schmeltz & Gamble, 2017).

The heightened frequency, intensity and duration of climate disasters which are intensified by higher temperatures, including drought, flood and cyclones, also have profound effects on mental health worldwide (Banholzer et al., 2014). As a consequence, populations experience increased risks of affective and anxiety disorders including post-traumatic stress, disturbed sleep, substance use and thoughts of suicide (Cianconi et al., 2020; Palinkas & Wong, 2020; Walinski et al., 2023). Severe cases of these are more prone to appear directly after disaster (Cianconi et al., 2020), but are also found to be long-lasting with delayed treatment-seeking (Kaniasty et al., 2020). Bushfires, for instance, can cause long-term psychological distress (Zhang et al., 2022), and populations that seek mental health treatment for related trauma may do so years later (Goldmann & Galea, 2014)

There is an increasing concern in Australia due to the rising climate, with a recorded warming of around 1.5°C since 1910 (Bureau of Meteorology & CSIRO, 2024). The population continues to experience growths in heatwave exposure, leading to higher likelihoods of continued and more extreme bushfires, drought and heat stress (Beggs et al., 2025). Through analyses on administrative data in Australian cities, hospital emergency presentations, admissions and mortality for most major presentation categories including mental health increases significantly on days of extreme heat (Nitschke et al., 2011; Vaneckova & Bambrick, 2013). Higher ambient temperatures also have an effect – the risk of mental health related hospital presentations increased on days with maximum temperatures above 36°C in Western Sydney (Ooi et al., 2024).

However, much of the current literature on temperature and mental health predominately relies on hospital emergency and admissions data to measure these outcomes (Massazza et al., 2022). While administrative data of this nature is effective in tracking serious outcomes from recorded diagnosis codes during emergency response, it is conversely more likely to miss the less severe cases and general burden of disease for mental health (Beggs et

al., 2022). There is also a lack of analyses using primary and community health care data (which would have greater sensitivity in capturing mental health effects) as supporting evidence for climate change impacts on the population (Massazza et al., 2022). Existing research on Australian climate disasters and mental health investigate impacts by surveying regional or vulnerable (e.g. in rural/agricultural areas) populations, rather than capturing health service use (Edwards et al., 2015; Leviston et al., 2023; Matthews et al., 2019). In Australia, primary health care data is collected through Medicare unit record claims data, which has nation-wide coverage of services claimed through public health for administrative purposes. Here, a broad range of services such as General Practitioner (GP) mental health care plans or Eating Disorder services are categorised under item numbers to fit the type of appointment a patient attends instead of using specific diagnosis codes.

Hence, this study intends to measure the impacts of climate change (i.e. extreme heat and climate disaster incidence) on mental health in Australia using publicly available primary health care data. Since treatment-seeking following disaster trauma can be delayed within 4-6 months of disaster (Elhai & Ford, 2009) with some seeking treatment only years later (Goldmann & Galea, 2014), an exploration of lagged effects will also be conducted. Of particular interest to the Australian climate are events involving extreme heat (including days of extreme heat, drought and wildfires) which are growing in frequency and severity (Beggs et al., 2025) and are connected to worsening mental health through heat stress (Liu et al., 2025). Other disasters, while still influenced by climate change like floods and cyclones, are very destructive and are not included in this study for sake of study scope and feasibility. Instead, this study will focus on disasters more closely related to very high temperatures, a more long-term concern for Australia.

Therefore, this project aims to investigate climate change associations with community mental health care service use (CMHSU). CMHSU in this paper is defined as the number of mental health-related primary and secondary (i.e. specialists like psychiatry) services provided to outpatients in Australia.

Specifically, we aim to investigate:

1) Associations between rising ambient temperature and CMHSU
2) Associations between extreme climate events (high heat days, heat waves, drought and bushfires) and CMHSU
3) Associations between climate events and total CMHSU during the COVID-19 emergency period
4) Associations between climate events and Telehealth CMHSU during the COVID-19 emergency period
5) Lagged effects of climate change events on CMHSU

Methods

Study Design

A retrospective longitudinal ecological study is conducted using publicly available primary health care, meteorological and disaster incidence data.

Ethics approval

This project uses publicly available datasets and does not involve human or animal subjects. Therefore, an ethical approval is not required.

Settings and Locations

This project uses datasets which are analysed at state-level across all Australian states and territories, using state-level monthly aggregates of CMHSU and climate exposures to reflect between-state differences. New South Wales (NSW) models include data from The Australian Capital Territory (ACT) as it is geographically enclosed in NSW, containing climate similarities and aligning with the Bureau of Meteorology's system of allocating station numbers (Bureau of Meteorology, 2011).

Population & Data Sources

Medicare Benefits Schedule

The Medicare Benefits Schedule (MBS) captures nation-wide use of Medicare-subsidised mental health services from general practitioners (GP), psychologists, psychiatrists and other allied health professionals. This cohort includes Australians enrolled in Medicare, as well as any eligible overseas visitors (Department of Health, Disability and Ageing, 2025). Primary and secondary care data is sourced from publicly available MBS Reports on covered professional attendances (*Services Australia - Statistics - Group Reports*, 2026). Service use is counted per capita for each state or territory, calculated using the number of processed attendances divided by the population enrolled in Medicare each month. It is noted that these monthly reports are restricted to services provided through public health. The main outcome variable for CMHSU is classified as mental health attendances claimed under MBS. As seen in Table 1, relevant mental health attendances are identified using MBS item categories.

Table 1

MBS Professional Attendance Categories Included for Analysis

Category Code	Description
A7 (Subgroup 9)	Prescribed medical practitioner mental health care
A8	Consultant Psychiatrist
A20	Mental health care including GP mental health care plans and focussed psychological strategies
A36	Eating Disorders (ED) services
A40 (Subgroups 3, 6, 9, 10, 19, 20, 21, 23, 25, 27 and 28)	Telehealth services for focussed psychological strategies, GP mental health treatment plans, with consultant psychiatrists and for ED management.
A41	Additional focussed psychological strategies
A42	Mental health planning of care recipients of a residential aged care facility

While the reporting of mental health services starts from 1994, the Better Access to Psychiatrists, Psychologists and General Practitioners through the Medicare Benefits Schedule initiative (Better Access) introduced a number of additional mental health service items available for rebate in November 2006 (Pirkis et al., 2022). Due to great increases in the utilisation of mental health services from Better Access (Pirkis et al., 2022), MBS data (and all other data sources) used for analysis commences from 1 November 2006 to 31 December 2023. This period is then split into two (November 2006 – February 2020, March 2020 – December 2023) for separate analyses. This is to account for the disruption of services, as well as the addition of Telehealth items from March 2020 in response to the COVID-19 pandemic (Department of Health, 2020).

Australian Climate Observations Reference Network - Surface Air Temperature

The Australian Climate Observations Reference Network - Surface Air Temperature (ACORN-SAT) datasets provide long-term homogenised surface air temperature records suitable for climate change detection across Australia (Trewin & Australia, 2018). As this study is an analysis on climate extremes, daily maximum temperature data for these 112 stations are obtained, categorised by state and merged manually. Given the frequency of MBS data is monthly, temperatures are also averaged over each month to match, representing monthly ambient max temperatures. Since the relationship between temperature and mental health outcomes may follow a U-shape (Hua et al., 2023), an additional predictor variable for temperature squared is included. Extreme heat days are identified by averaging daily station records across each state, counting the number of days with temperatures above the 95th percentile, per Adams and colleagues (2022).

ACORN-SAT data is mostly complete – 1.45% of daily data within the study period is missing and most weather stations have less than 5% of records missing. After 1971, there are at most 2 sites missing for any given year (Trewin & Australia, 2018).

Emergency Events Database (EM-DAT)

Historical records of Australian disasters are sourced from the EM-DAT International Database (Delforge et al., 2025) with heatwaves, wildfires and drought classified as climate-relevant disasters (Xu et al., 2023). Each disaster is defined as occurring for the month if at least one day is within its start-end date range.

Location information from this dataset is parsed from a free text field listing affected states, cities and suburbs. As climate disasters may not affect the whole state it is in, some exclusions were made according to which areas/proportion of a states was exposed. No drought or heatwave events were excluded as they tend to have far-reaching effects that can spread to much of the state if not country. This is reflected in the EM-DAT database as impacted locations are recorded more generally and across larger areas of a state. On the other hand, wildfires can have more local impacts. Hence, wildfire observations are only included when major cities or a larger proportion of the state are recorded. Observations limited to only smaller towns or sparsely populated suburbs are excluded (e.g. the 2011 Margaret River fire; Australian Disaster Resilience Knowledge Hub, 2011), unless the database specifies a broader regional-level location, main city or state as location information. Wildfires with the main cause attributed to industrial accidents such as the Hazelwood open cut mine fire (Australian Disaster

Resilience Knowledge Hub, 2014) were also excluded, as these do not necessarily represent climate change-related disasters.

Exposure and Outcome Variables

A summary of the covariates and outcomes used for modelling are outlined in Table 2.

Table 2

Definitions for Predictor and Outcome Variables

Data source	Locations	Variable name	Description	Coding
MBS	State or territory	*MBS Total*	Mental health-related primary/secondary services covered by Medicare	Numeric, count per capita (100,000)
		MBS TH	Mental health-related primary/secondary services provided over Telehealth	Numeric, count per capita (100,000)
ACORN-SAT	Individual weather stations (*n* = 112)	*Monthly temperature*	Monthly surface max temperature to represent ambient temperatures experienced by the population	Numeric (°C)
		High temp count	Number of days within a month where the max temperature is above the state's 95th percentile	Numeric
EM-DAT	Some or all of districts, cities and states exposed (in a free text field)	*Heatwave flag*	Flags for occurrence of heatwave within the month	Binary, 0 – No heatwave 1 – Heatwave
		Drought flag	Flags for occurrence of drought within the month	Binary, 0 – No drought 1 – Drought
		Wildfire flag	Flags for occurrence of wildfire within the month	Binary, 0 – No wildfire 1 – Wildfire

Monthly temperature, high temperature counts and heatwave flag variables are included to capture different dimensions of heat exposure. *Monthly average temperature* is used to represent fluctuations in ambient temperatures and to align with metrics used to make predictions from future climate projections. This also allows the inclusion of a squared temperature term in case of potential non-linear relationships with mental health outcomes. However, because these averages smooth over daily variation, analyses will be unable to detect short term peaks and large fluctuations in the weather will dilute effects. Hence, *high temp count* is incorporated as a second predictor to bring more focus on the effects of extreme heat from global warming and avoid the influence of colder days that can obscure monthly temperatures. In this study, high heat days are treated as a climate disaster. Finally, *heatwave flags* represent nationally recognised periods of sustained, significantly elevated temperatures as recorded in the EM-DAT database. These are relevant for parsing the effects of cumulative heat and physiological stress.

Statistical Methods

All data manipulation and analyses will be performed using R statistical software version 4.5.0 (R Core Team, 2025). R packages used for analyses are outlined under Table 3.

Preliminary descriptive statistics are carried out for a crude analysis of CMHSU across states and territories. An interrupted time series analysis is also conducted to check for significant impacts from the introduction of new MBS items during each study window.

Table 3

R Packages Used for Data Cleaning and Analysis

Package name	Use for Cleaning/Analysis
tidyverse	Data cleaning, management, restructuring and visualisation
janitor	Quick cleaning of variable names
stringr	Parsing information from free-text fields
lubridate	Better management of date objects
forecast	ARIMA modelling and visualising data seasonality
astsa	Interrupted time series analysis of MBS policy changes
mice	Multiple imputation for missing data
miceadds	Additional methods for multiple imputation. Specifically, for predictive mean matching of multilevel data
imputeTS	Testing time-series imputation of missing data
stats	Core R stats package for generalised linear modelling
tsModel	Harmonic function to generate Fourier terms

Note. (Firke, 2024; Grolemund & Wickham, 2011; Hyndman et al., 2025; Moritz & Bartz-Beielstein, 2017; Peng, 2024; R Core Team, 2025; Robitzsch & Grund, 2026; Stoffer, 2025; van Buuren & Groothuis-Oudshoorn, 2011; Wickham, 2023; Wickham et al., 2019)

Imputation of Missing Values

While the overall proportion of missing temperature data remains very low (1.45%), two notable weather sites had many extended periods of missing observations (i.e. Boulia Airport at 27.4% missing, Gunnedah Resource Centre at 20.85% missing). Both sites were removed from final analysis, bringing down the total proportion of missing data to 1.04% and with no large impact expected on overall results.

The remaining of these values were addressed using imputation methods. A series of imputation strategies were compared, including a complete-case analysis, time-series imputation using *imputeTS* (Moritz & Bartz-Beielstein, 2017), and multiple imputation with several methods (specifically '*pmm', '2l.norm'* and *'2l.pmm'*). Even with little deviation between non-imputed and imputed datasets after averaging over state and month, multiple imputation was selected to reduce the risk of Missing Not At Random (MNAR) bias and preferred over time-series imputation for the pooling of results across multiply imputed datasets using Rubin's Rules. Given the clustering of temperatures within and variation between weather stations, two-level ('*2l*') methods are used to appropriately model this multilevel structure. The *'2l.norm'* method, while computationally efficient, assumes normality and tail ends of imputed values can surpass observed temperature minimums or maximums. For this reason, '*2l.pmm*', a

multilevel predictive mean matching method, was ultimately chosen as it better preserves the distributional shape of observed data and avoids exceeding thresholds.

To preserve the temporal structure of time-series data, 1 day lag and lead variables were created prior to imputation. Additional variables with deterministic transformations required for later analysis (e.g. average daily and monthly temperatures, state 95th percentile thresholds and flags for extreme heat days) were also created for passive imputation following the recommendations of White et al. (2011). The two-level predictive mean matching method is used with five multiply imputed datasets (m = 5) sufficient for <5% missingness (White et al., 2011). This process was run for ten iterations (maxit = 10), which was adequate for achieving stable and well-mixed trace plots indicating convergence (van Buuren, 2018).

Modelling and Analysis

Statistical analyses are performed separately for each state/territory through a time-series regression as described by Bhaskaran and colleagues (2013). A quasi-poisson regression model is fit to the MBS data using CMHSU counts per capita. Seasonality and long-term trends not attributable to the main exposure are controlled for using Fourier paired sine/cosine functions and a linear function respectively. An exposure-outcome model is then fit using these seasonality and trend adjustments for all aims. Since ambient temperature can have a U-shaped relationship with health outcomes like mortality and symptoms of depression (Armstrong, 2006; Hua et al., 2023; Kinney et al., 2008), a quadratic term is tested when modelling the effects of overall changes in maximum temperature. However, this quadratic term is shown to have very low estimates and standard errors across Australia with no significant contribution to the model and is thus removed from final analyses.

For Aims involving climate disasters, the effect of each climate type will be modelled separately as a binary variable (Table 4). As the EM-DAT database shows no records of drought or heatwave during the COVID-19 period, these covariates were also removed from the final model for Aims 3 and 4. 1-3 month and 12 month lags are chosen to be investigated in Aim 5 due to climate disasters having both medium and long-term impacts on mental health (Elhai & Ford, 2009; Goldmann & Galea, 2014; Zhang et al., 2022).

Table 4

Study Period and Modelled Variables for Aims

Aim	Time Period	Outcome (per capita)	Modelled variables
1 and 2	Nov 2006 - Feb 2020	Total MBS count	Monthly max temperature, high heat days, binary flags for heat wave, drought and bushfire
3	Mar 2020 - Dec 2023	Total MBS count	Monthly max temperature, high heat days, binary flags for bushfire
4	Mar 2020 - Dec 2023	MBS count for Telehealth services	Monthly max temperature, high heat days, binary flags for bushfire
5	Nov 2006 - Feb 2020	Total MBS count for 1-3 and 12 month lags	High heat days, binary flags for heat wave, drought and bushfire

To evaluate the goodness-of-fit of these models, the proportion of explained deviance is calculated and residual deviances compared. To check for model assumptions and sensitivities, diagnostic plots on deviance residuals are constructed and a sensitivity analysis conducted using 3 to 5 sine/cosine pairs conducted.

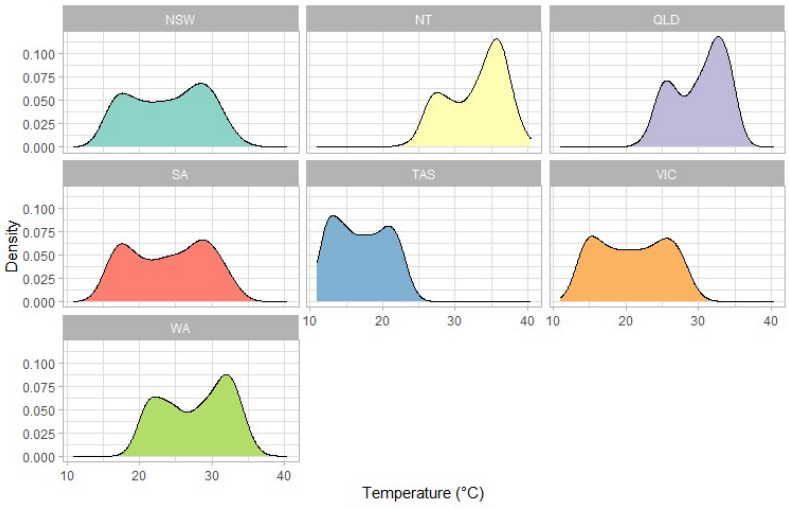
Results

Descriptive Statistics

In the Nov 2006 - Dec 2023 study period, the average max temperature in Australia is estimated at 25.2°C with a standard error of 0.177. The distributions of these temperatures differ between states (Figure 1), with southern states like Tasmania (TAS) and Victoria (VIC) showing the lowest average temperatures (17.0°C and 20.8°C respectively) and northern states with the highest (e.g. Northern Territory at 32.9°C).

Figure 1

Max Temperature Distribution by State



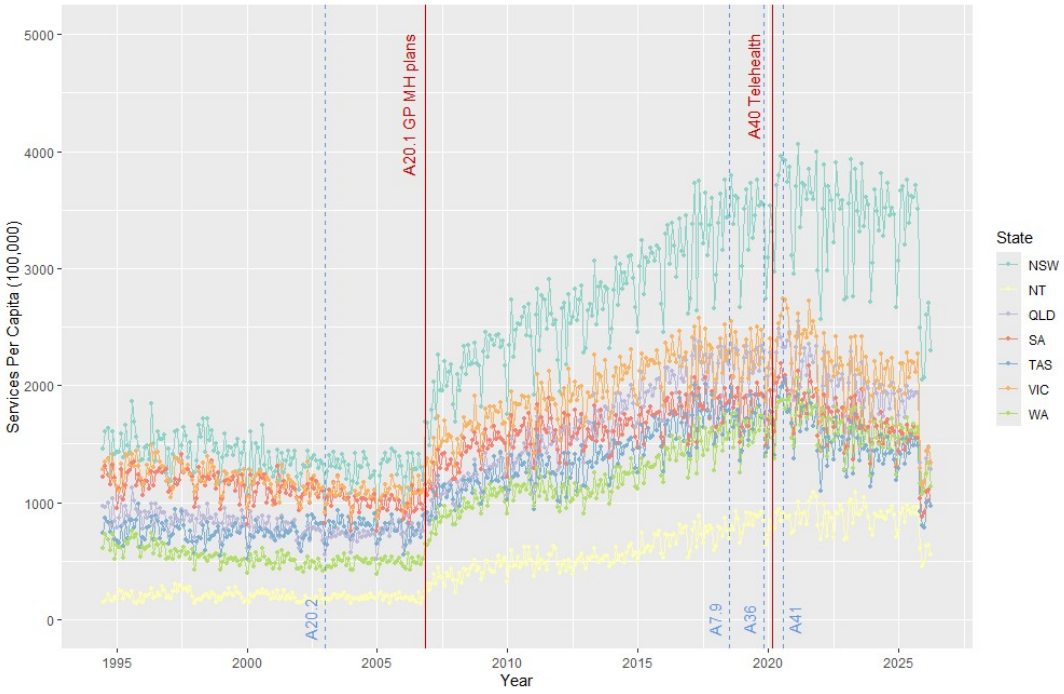
Note. NSW = New South Wales, NT = Northern Territory, QLD = Queensland, SA = South Australia, TAS = Tasmania, VIC = Victoria, WA = Western Australia.

CMHSU also differs by state. As seen in Figure 2, the Northern Territory (NT) consistently records the lowest number of mental health services, with a monthly median of 613 per 100,000 Medicare-eligible population. NSW regularly provides the most Medicare subsidised mental health services with a median of 2861 (IQR 2456 – 3434) total and 756 Telehealth services per capita during the Mar 2020 – 2023 pandemic period. In the prior Nov 2006 – Feb 2020 period, there is a trend of increasing service use for every state and territory, the fastest growth attributable to time being in NT at an average growth of 6.72% every year.

For climate disasters, wildfires were the most common nation-wide at a prevalence of 3.1%, while drought and heatwaves happened 2.6% and 1.4% of the time respectively. Queensland (QLD) recorded the most drought months at a rate of 4.9%, whereas wildfires occurred the most for Western Australia (4.4%). During the COVID-19 emergency period, no droughts or heatwaves were recorded in the EM-DAT database, so modelling in this period (Aims 3-4) will not involve these variables as predictors.

Figure 2

MBS Subsidised Mental Health Services



Note. This figure depicts the rate of all mental health related services subsidised by Medicare per capita in Australia. This includes the introduction of new items as depicted by the labelled vertical lines. The red lines for A20.1 GP MH plans and A40 Telehealth indicate important time points which made significant changes to future trends and are accounted for in this study.

Interrupted Time Series Analysis

An interrupted time series analysis was performed for the introduction of prescribed practitioner care services (A7.9) in MBS mental health data from November 2019 onwards. Autoregressive Integrated Moving Average (ARIMA) modelling was selected as the most appropriate approach given the seasonal structure of this data (Hyndman & Khandakar, 2008). The analysis found no significant change in monthly rate of services following the addition of this item (p = .13), indicating this policy change does not confound main results.

Additional new item codes introduced during the study period (A36 and A41; depicted in Figure 2) were not analysed further because recorded values were very low (<10).

Modelling Diagnostics

Diagnostic checks were conducted for the baseline trend and seasonality models to ensure adequate fit before adding climate change predictors. Pooled dispersion parameters decreased as the number of sine-cosine pairs increased from 3 to 5. All parameter values remain greater than 1 (Table 5), confirming overdispersion and supporting the use of a quasi-Poisson model. Model fit also improved modestly with added Fourier terms as the proportions of total deviance explained increased by a few percent. Autocorrelation function (ACF) plots show models with 4-5 Fourier pairs reduced autocorrelation relative to 3-pair models, though

higher-order lags still exhibited some dependence. Based on these diagnostics, 5 sine-cosine pairs were selected for all final models.

Queensland and Western Australia (WA) in particular displayed pronounced and significant autocorrelation. Residual over time plots further indicate poorer fit for QLD and WA, with QLD showing a reverse-U residual pattern and WA an oscillating trend. ACF and residual plots for each state can be accessed in Appendix A.

Table 5

Baseline Model Assumption Check and Diagnostics

		State						
	Fourier terms	NSW	NT	QLD	SA	TAS	VIC	WA
Dispersion parameters	3	14.8	7.2	10.2	9.4	10.4	10.4	8.8
	4	9.3	6.0	8.1	6.8	8.2	8.1	6.8
	5	8.4	5.9	7.5	6.4	7.5	6.3	6.1
% Deviance explained (Model / Null Deviance)	3	86.7%	85.7%	90.2%	69.4%	81.4%	80.2%	85.4%
	4	91.7%	88.1%	92.3%	78.2%	85.6%	84.8%	88.9%
	5	92.7%	88.6%	93.0%	79.7%	87.0%	88.3%	90.2%

Aims 1-2: Temperatures and Climate Disasters pre-Covid

Table 6 presents the exponentiated model estimates (Risk Ratios) and corresponding p-values for each state during the pre-COVID period, summarising climate-related predictors of CMHSU. As expected, time is a significant variable across Australia, showing a general increasing trend for MBS-funded mental health service use over the years. Overall, changes in ambient max temperatures, the occurrence of heatwaves and the number of high temperature days in a month were found to have no significant effect on CMHSU. There is one exception for high temperatures in South Australia (SA), such that for every increase in the number of high temperature days within a month, the rate of CMHSU is estimated to decrease by 0.83% ($p = .014$).

By contrast, wildfires and drought demonstrated significant negative effects on CMHSU in several states. Drought was associated with reduced service use in NT, SA and WA with the WA model showing a particularly strong effect. At a risk ratio of 0.769 [95% CI: 0.613 - 0.882], the months in which a drought occurs in WA on average, predicts a decrease in mental health services to 76.9% of typical levels. On the other hand, wildfires were found to be negative predictors of CMHSU in NSW ($p = .004$) and QLD ($p < .0001$), and borderline significant for Victoria ($p = .05$). As illustrated in Figure 3 and Figure 4, effect sizes and confidence intervals varied substantially across states, highlighting that there is still a cross-state heterogeneity in climate-related impacts on mental health treatment-seeking.

Table 6

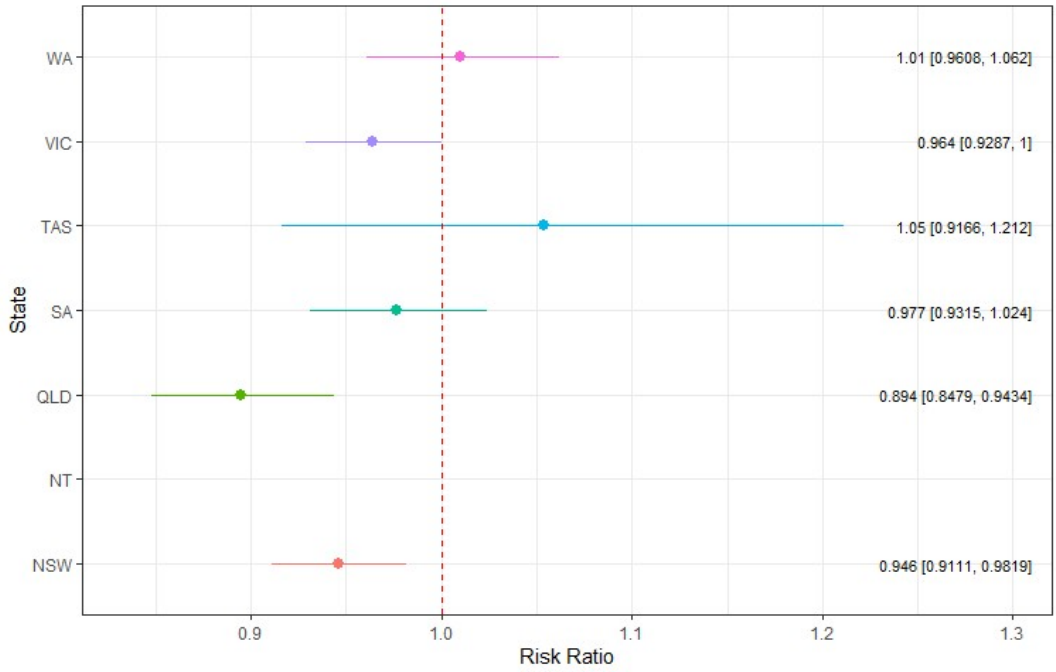
Climate Change Related Model Estimates by State for Nov 2006 – Feb 2020

		State						
Covariate		NSW	NT	QLD	SA	TAS	VIC	WA
Time	*Risk Ratio*	**1.0038**	**1.0056**	**1.0051**	**1.0021**	**1.0035**	**1.0029**	**1.0042**
	p value	**1.45×10^{-73}***	**1.11×10^{-61}***	**3.78×10^{-76}***	**7.68×10^{-41}***	**3.65×10^{-58}***	**3.74×10^{-60}***	**8.86×10^{-67}***
Monthly Temperature	*Risk Ratio*	0.9980	1.0008	1.0085	0.9978	0.9987	1.0003	0.9984
	p value	0.6643	0.9153	0.1756	0.6215	0.8955	0.9559	0.8255
High temperature Count	*Risk Ratio*	1.0029	0.9937	0.9986	**0.9917**	1.0041	0.9947	0.9962
	p value	0.2529	0.0967	0.5434	**0.0142***	0.4020	0.2061	0.2089
Heatwave	*Risk Ratio*	0.9344	1.1405	0.9221	0.9863	0.9496	0.9831	0.9911
	p value	0.1245	0.1355	0.0770	0.7631	0.2208	0.5826	0.8299
Wildfire	*Risk Ratio*	**0.9458**		**0.8945**	0.9767	1.0538	0.9637	1.0103
	p value	**0.0038***		**0.0001***	0.3280	0.4587	0.0502	0.6877
Drought	*Risk Ratio*	1.0017	**0.8067**	0.9696	**0.8333**	0.8781	0.9842	**0.7694**
	p value	0.9252	**0.0450***	0.1280	**0.0007***	0.0861	0.3839	**0.0002***

*$p < .05$

Figure 3

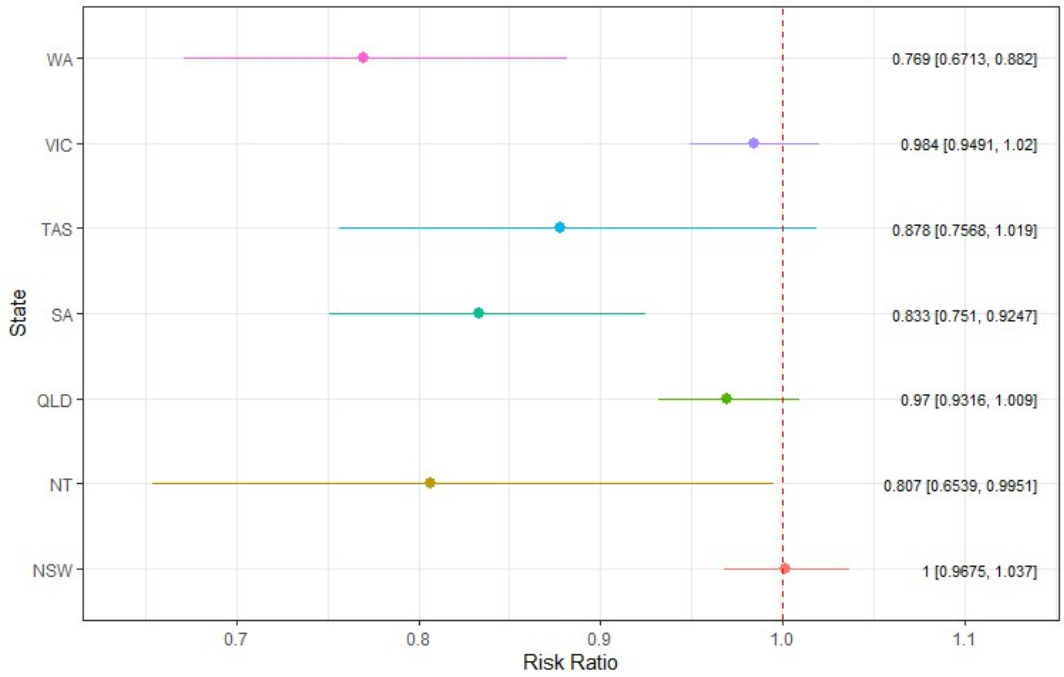
Confidence Intervals for Wildfire Effects in pre-COVID Models



Note. Intervals are shown at the 95% confidence level.

Figure 4

Confidence Intervals for Drought Effects in pre-COVID Models



Note. Intervals are shown at the 95% confidence level.

Aims 3-4: Temperatures and Climate Disasters during Covid-19

Modelling for these aims focus on fluctuations in ambient temperatures, high-temperature days and wildfires as explanatory variables. None of the investigated climate exposures had a significant effect on CMHSU during this period, a pattern which remained consistent when modelling total MBS mental health services or isolating to just Telehealth services. The full table of results for this can be found in Appendix B.

Aim 5: Lagged Effects of Climate Disasters

Table 7 summarises the Risk Ratios for estimated climate disaster effects, providing evidence that wildfires and drought exert lagged negative effects on CMHSU, most commonly within the first few months from exposure. In NSW and QLD, wildfires were associated with reduced CMHSU immediately after the event and throughout the subsequent three months, indicating a short-term decline in treatment-seeking. WA exhibited a similar pattern for drought, and with greater effect sizes, particularly in months closely following exposure (Figure 5). Drought also produced lagged impacts in QLD, though these effects emerged only after the month of drought, rather than during it. No lagged associations were detected for NT or Tasmania, demonstrating geographic variability in disaster-related impacts. Heatwaves showed little meaningful lagged effects, except for in QLD from 2 months after exposure.

Table 7

Lagged Climate Disaster Model Estimates by State for Nov 2006 – Feb 2020

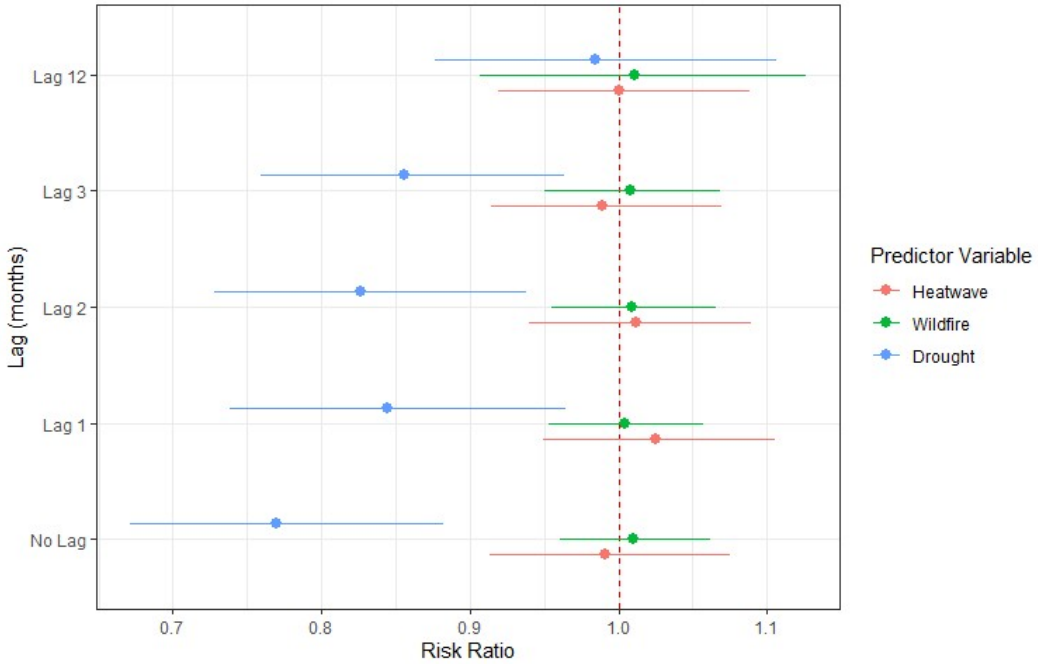
Model	Covariate		State NSW	NT	QLD	SA	TAS	VIC	WA
Immediate Effect	Heatwave	*Risk Ratio*	0.9344	1.1405	0.9221	0.9863	0.9496	0.9831	0.9911
		p value	0.1245	0.1355	0.0770	0.7631	0.2208	0.5826	0.8299
	Wildfire	*Risk Ratio*	**0.9458**		**0.8945**	0.9767	1.0538	0.9637	1.0103
		p value	**0.0038***		**0.0001***	0.3280	0.4587	0.0502	0.6877
	Drought	*Risk Ratio*	1.0017	**0.8067**	0.9696	**0.8333**	0.8781	0.9842	**0.7694**
		p value	0.9252	**0.0450***	0.1280	**0.0007***	0.0861	0.3839	**0.0002***
Lag - 1 month	Heatwave	*Risk Ratio*	0.9684	1.1040	0.9553	0.9939	0.9706	0.9861	1.0245
		p value	0.4318	0.2349	0.2770	0.8930	0.4582	0.6301	0.5306
	Wildfire	*Risk Ratio*	**0.9359**		**0.8766**	0.9723	0.9796	**0.9496**	1.0038
		p value	**0.0010***		**1.28×10^{-5}***	0.2953	0.7592	**0.0079***	0.8869
	Drought	*Risk Ratio*	1.0011	0.9192	**0.9581**	0.9181	0.9840	0.9719	**0.8439**
		p value	0.9498	0.4218	**0.0333***	0.1216	0.8305	0.1146	**0.0132***
Lag - 2 month	Heatwave	*Risk Ratio*	0.9474	1.0956	**0.9160**	0.9737	0.9567	1.0036	1.0115
		p value	0.1620	0.2569	**0.0340***	0.5332	0.2551	0.9013	0.7594
	Wildfire	*Risk Ratio*	**0.9212**		**0.9033**	0.9731	0.8981	0.9624	1.0085
		p value	**0.0001***		**0.0021***	0.3322	0.1100	0.0570	0.7624
	Drought	*Risk Ratio*	0.9984	0.8596	**0.9566**	**0.8632**	0.9449	**0.9600**	**0.8260**
		p value	0.9218	0.1490	**0.0279***	**0.0059***	0.4574	**0.0248***	**0.0034***
Lag - 3 month	Heatwave	*Risk Ratio*	0.9526	0.9978	**0.8936**	0.9654	0.9749	0.9989	0.9889
		p value	0.2349	0.9826	**0.0084***	0.4139	0.5140	0.9701	0.7787
	Wildfire	*Risk Ratio*	**0.9551**		**0.8933**	0.9543	1.0300	0.9710	1.0076
		p value	**0.0424***		**0.0018***	0.1296	0.6534	0.1533	0.8007
	Drought	*Risk Ratio*	0.9982	0.8898	**0.9610**	**0.8373**	0.8789	**0.9613**	**0.8552**
		p value	0.9184	0.2258	**0.0440***	**0.0004***	0.0760	**0.0268***	**0.0106***
Lag - 12 month	Heatwave	*Risk Ratio*	0.9230	0.9056	**0.9058**	0.9621	0.9328	**0.9394**	1.0007

	P value	0.0581	0.3562	**0.0227***	0.3699	0.0952	**0.0330***	0.9862
Wildfire	*Risk Ratio*	1.0353			1.0015	1.0263	0.9903	1.0102
	P value	0.2502			0.9799	0.7022	0.7510	0.8524
Drought	*Risk Ratio*	0.9671	1.0141	**0.9283**	0.9181	1.0131	**0.9529**	0.9847
	P value	0.0542	0.8783	**0.0002***	0.0653	0.8501	**0.0065***	0.7940

*$p < .05$.

Figure 5

Confidence Intervals for Disaster Effects in Lagged Models



Note. Intervals are shown at the 95% confidence level.

Discussion

High Temperature Effects

Rising ambient temperatures and the frequency of extreme heat days showed little to no significant association with CMHSU in this study. Across most states, neither of these variables were statisitically related to mental health service use. Only NSW and QLD displayed a small effect in their one-month lag models, where each additional high temperature day above average in the preceding month corresponded to an estimated 0.4% increase in mental health service utilisation. Heatwaves, another closely related exposure, also showed minimal significance for CMHSU.

These results conflict with previous findings of existing Australian evidence. Prior studies have reported clear associations between deteriorating mental health and rising ambient temperatures (Ooi et al., 2024), days of extreme heat and heatwaves (Nitschke et al., 2011; Vaneckova & Bambrick, 2013; Xu et al., 2023). The discrepancy may reflect limitations inherent to monthly time-series data, which can mask short-term events of extreme heat that exert psychological stress on the population. Although this study attempted to mediate this by incorporating both monthly max temperature means and counts for the number of days in which temperatures rise above their states' 95th percentile, extreme temperature episodes still notably unfold over shorter periods not captured within monthly blocks. Differences in health-seeking behaviour may also contribute. Australians may rely on primary or secondary care differently from emergency departments, which provide more critical care for acute mental health issues and can miss subtle or longer-term effects on the population. A recent pre-print using weekly MBS data found mental health care increased during acute heatwaves, but showed little association with longer-term heat exposure (Chomik et al., 2025). It did not account for other climate-related disasters such as drought or bushfires, as investigated by this present study.

Climate Disasters

In contrast, this study found evidence that climate disasters, particularly wildfires and drought, were associated with decreases in mental health service use following exposure. Although patterns were not uniform across states, significant negative associations emerged for wildfires in NSW and QLD, and drought in NT, SA and WA. State-specific drought effects may reflect land use composition and underlying climatic conditions. SA is Australia's driest state (Tan, 2024), while WA's interior and north and NT's southern regions are very arid, experiencing similar pressures (Lightner, Barb, 2020; Northern Territory Government, 2022). Climate variability including drought is a well-established risk factor for farm-worker mental health (Tong et al., 2024), and has the most prominent impact in the south-west and south-east regions of Australia's mainland (*Snapshot of Australian Agriculture 2026*, 2026). QLD has the largest proportion of agricultural land, particularly grazing land, while WA and NSW contain the largest areas of cropland. As such, these agricultural and climatic characteristics may shape both exposure intensity and detected effects on mental health service use during disaster periods.

These findings, diverge from the current literature, which links climate disasters to worsening mental health (Cianconi et al., 2020; Palinkas & Wong, 2020; Tong et al., 2024; Walinski et al., 2023). Woods et al. (2026), on the other hand, discusses that the majority of

populations exposed to natural disasters do not seek mental health support, but proportions tend to increase with rising degrees of exposure. In the short term after natural disasters, the rate non-fatal suicidal behaviours also decreases, with increases in suicidal ideation occurring after delay (Kõlves et al., 2013). There are several other possible explanations for this – community health service availability may decline immediately following disasters due to damaged infrastructure, disruptions to accessibility, or the prioritisation of immediate physical needs. Patients may also seek emergency care over primary mental health services during crises.

COVID-19 Pandemic Period

Conducting similar analyses during the COVID-19 Emergency period found no significant predictors of CMHSU from high temperatures nor disasters – a pattern that held when focusing on just Telehealth services. This likely reflects the constraints of a shorter study window, during which seasonal patterns in MBS data were less clearly expressed. Along with substantial variation in random effects and only one disaster type (wildfires) observed, model estimations become more challenging. Thus, disruptions to daily living and the health care system throughout the pandemic may have obscured any underlying climate-related influences on mental health service use.

Lagged Effects

Climate Change related disasters also showed lagged effects on mental health treatment seeking, with wildfires and drought emerging as exposures with the most persistent negative associations across models. These disasters appear to exert longer lasting disruptions to CMHSU, while heatwaves and high heat days – being less destructive on the physical environment and typically shorter in duration – did not show comparable effects. Results also differed by state: NT and TAS displayed mostly non-significant predictors, while QLD's lagged models showed several lagged covariates with significant negative associations, suggesting prolonged reductions in service use after disaster exposure. While difficult to quantify the degree, these state differences can be attributed to the already low usage rate of MBS-subsidised mental health services in the Northern Territory and Tasmania.

Overall, this pattern aligns partially with broader research on delayed disaster impacts (Elhai & Ford, 2009; Goldmann & Galea, 2014), which demonstrates mental health consequences often unfold over extended periods, though with the opposite direction of effect. Temporary protective factors such as community cohesion and short-term social supports may buffer immediate impacts delaying treatment-seeking until supports recede (Kaniasty et al., 2020). But in the medium-term, the uptake of specialist mental health services is poor, public eye on affected communities and reported community unity is short-lasting (Kaniasty et al., 2020; McFarlane & Williams, 2012). Evidence from meta-analyses reinforces the potential for long term distress with bushfires, where psychological issues remain prevalent at 14% 2-4 years later (Zhang et al., 2022). This context suggests that while climate events may not immediately translate into changes in CMHSU, especially given the shorter 1-3 month and 12-month lags in this study, their impacts on mental health treatment seeking can persist and evolve over time.

Limitations

The key limitations of this study relate to the publicly available datasets used for modelling and analysis. MBS data captures only services delivered through Australia's public health system and therefore does not represent the full spectrum of CMHSU. Australians also receive support through private providers, or are covered by worker's compensation or a third party, none of which would be reflected in MBS item counts. In addition, individual mental health treatment services are capped at ten subsidised attendances per patient per year, meaning service use beyond this threshold is not observable. As a result, the dataset provides an incomplete picture of CMHSU. Furthermore, MBS item codes only allow for a broad categorisation of attendances such that specific diagnoses such as anxiety and depression are not recorded, limiting the ability to examine more specific responses and impacts.

Limitations also arise from missingness in the ACORN-SAT dataset. While these gaps can often occur randomly with damaged instruments, incorrect recordings or manual errors, an assessment of the missing data suggested the potential for MNAR mechanisms. Extended periods of missing data may coincide with extreme weather events, raising the possibility of systematic missingness related to the disasters examined in this study. Although overall missingness is very low and expected to have minimal influence on results, especially with multiple imputation, these patterns nonetheless raise uncertainty into the temperature measures used.

Methodological constraints further shape our findings. This study relies on data aggregated monthly and on the state level, a very broad scale that can obscure short-term effects. Temperatures can sometimes vary significantly between weather stations even in the same state (e.g. Cabramurra SMHEA AWS weather station in the Snowy Mountains). Climate disasters are often geographically specific. Even with the filtering of wildfire disasters affecting smaller proportions of a state, the integration of EM-DAT disaster records into analyses remain coarse. This makes it difficult to detect impacts that may be substantial locally but diluted when averaged across an entire state. Seasonal MBS patterns were modelled using sine and cosine functions, which assume consistent seasonality across all years. This can be problematic when unobserved confounders such as policy changes or amendments to MBS items alter seasonality within the study period. Some significant results, like in Queensland's disaster models, may reflect poorer model fit rather than true underlying associations. Residual diagnostics showed a slight trend in residual variation, suggesting that relationships between variables may not be fully linear and that more flexible modelling approaches could capture these dynamics more accurately.

Future Directions

Future research should draw on more detailed datasets to better investigate fluctuations in mental health service outcomes. Analyses stratified by demographics, vulnerable groups and minorities, or additional disaster types (e.g. floods or cyclones) would deepen understanding of differential impacts. Using smaller geographic units such as Public Health Networks or Statistical Area Level 3 would more validly track temperatures, refine disaster exposure indicators, and reveal impacts masked at the state level.

Conclusion

While mental health is largely understood to be negatively impacted by climate change effects, previous Australian studies rarely investigate these impacts on the population's primary and secondary health care use. As an exploratory investigation using national public administrative data, this project addresses this gap by modelling community mental health care attendances using high temperatures and climate disaster incidences. These health services play a central role in early intervention, recovery and chronic disease management, and help reduce pressure on emergency departments - an important consideration as climate change impacts continue and intensify. This study provides evidence that mental health treatment-seeking can decrease or be delayed following exposure to climate disasters, particularly drought and wildfire. It reveals the need for considerations on what the availability of community health services looks like following disasters and whether service use is more dependent on time or accessibility. The modelling framework developed here also offers a foundation for analysing similar datasets. This enables future work to track climate change burdens on Australia's public health system or inform projections on how it may function under these stresses moving forward.

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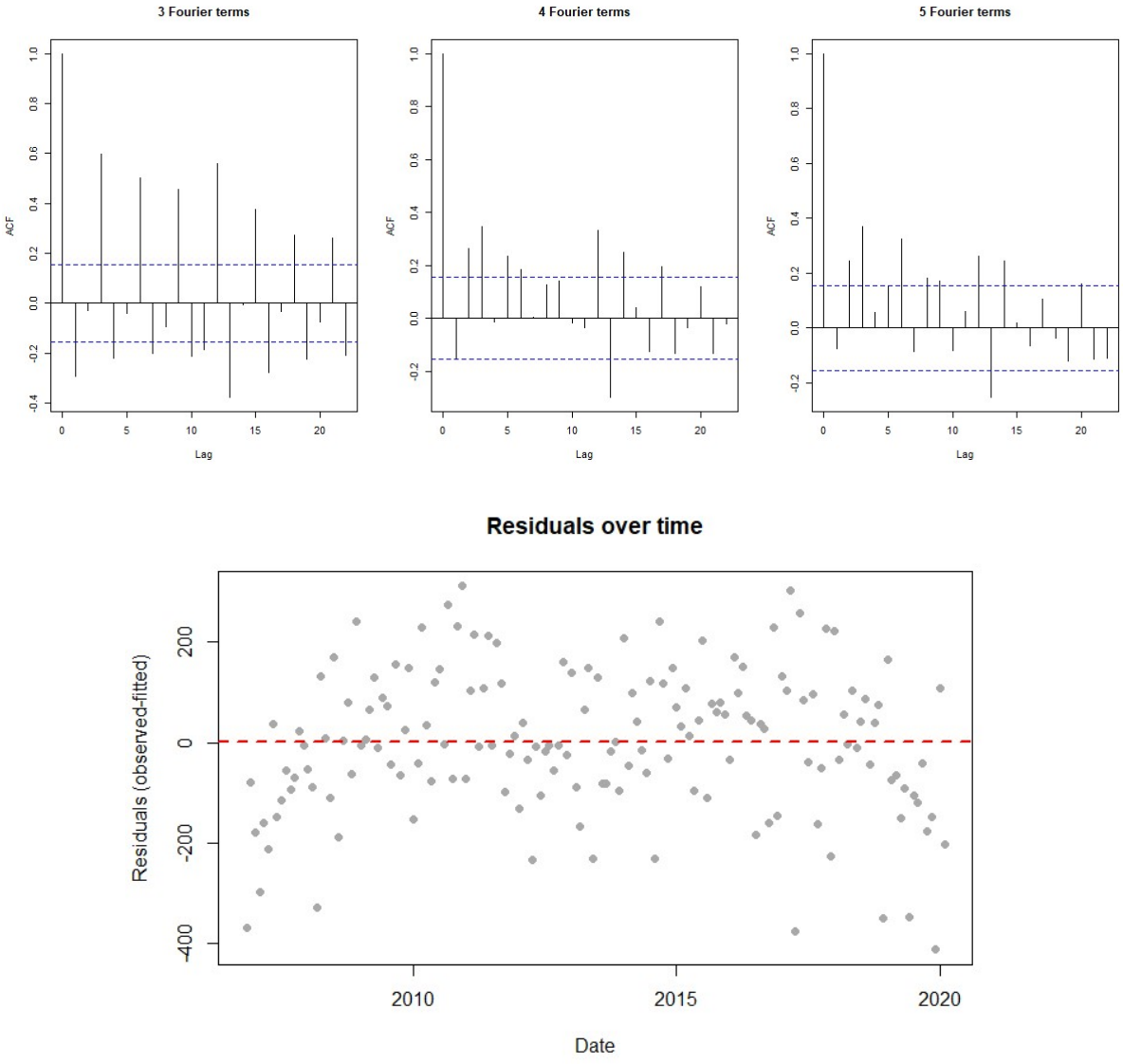
Zhang, Y., Workman, A., Russell, M. A., Williamson, M., Pan, H., & Reifels, L. (2022). The long-term impact of bushfires on the mental health of Australians: A systematic review and meta-analysis. *European Journal of Psychotraumatology*, *13*(1), 2087980. https://doi.org/10.1080/20008198.2022.2087980

Appendix A

Residual vs Fitted Values and ACF Plots for Baseline Models

Figure A1

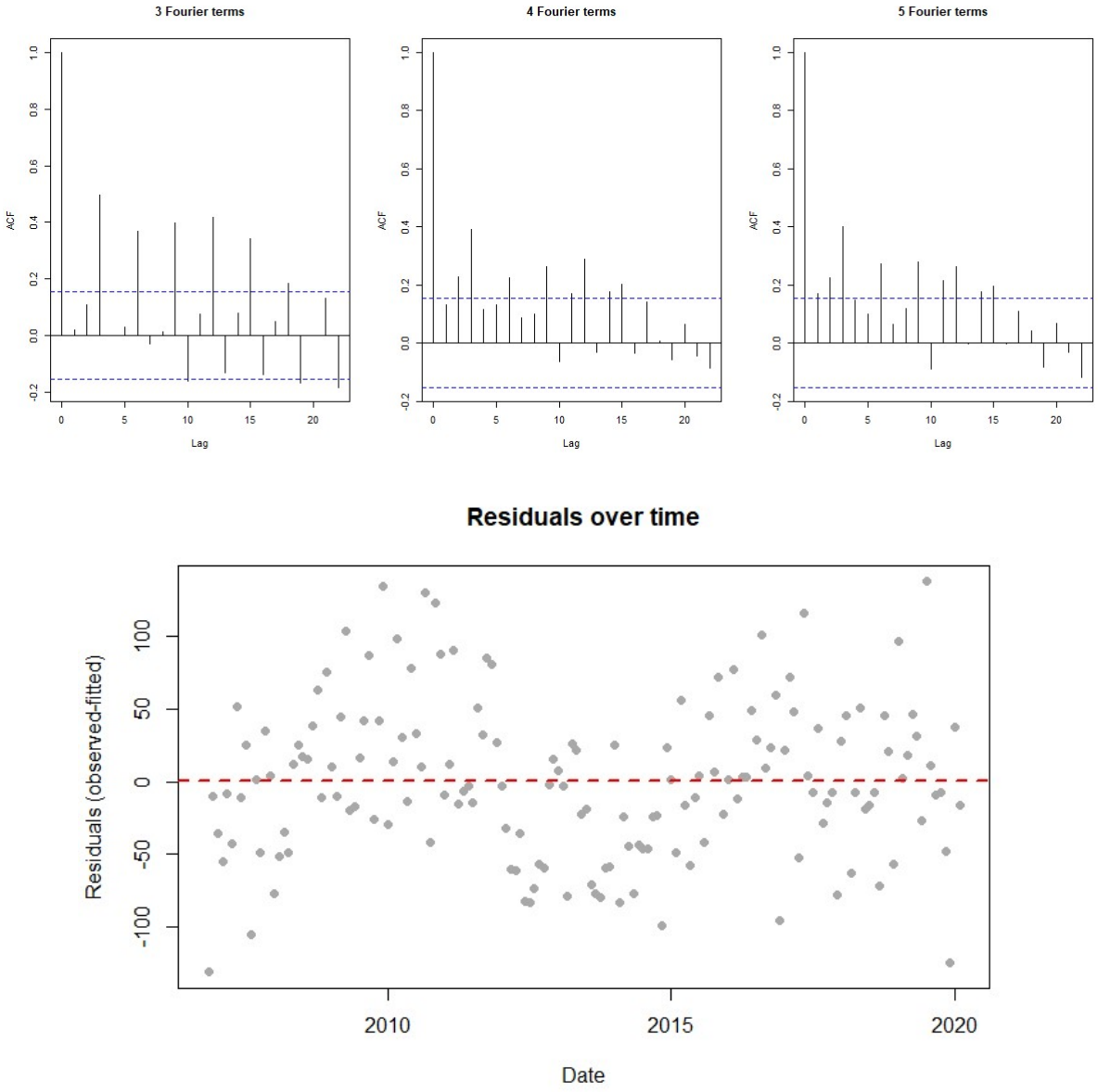
Diagnostic Plots for New South Wales Baseline Model



Note. The residuals over time plot is constructed from a 5 sine-cosine pair model.

Figure A2

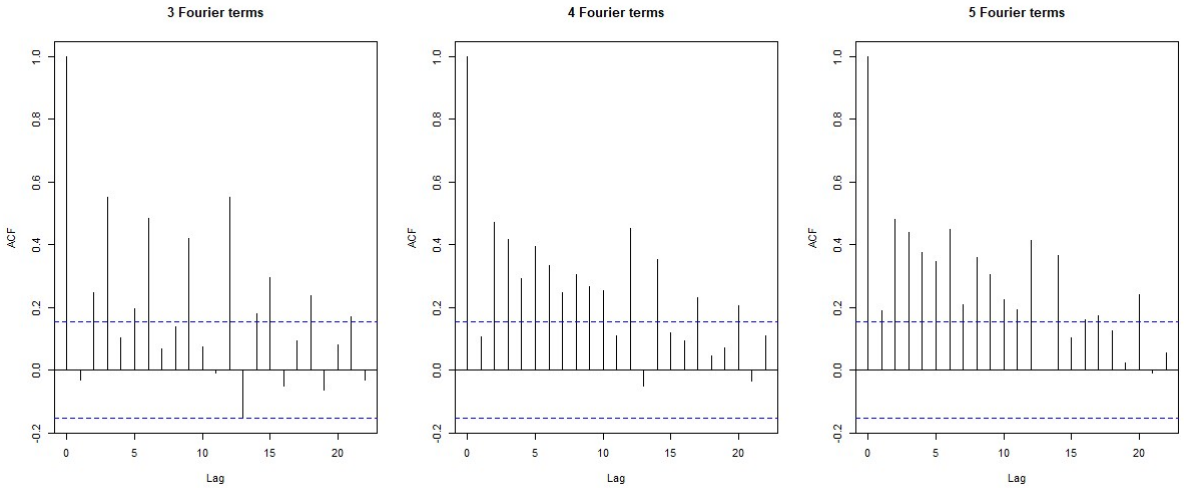
Diagnostic Plots for Northern Territory Baseline Model

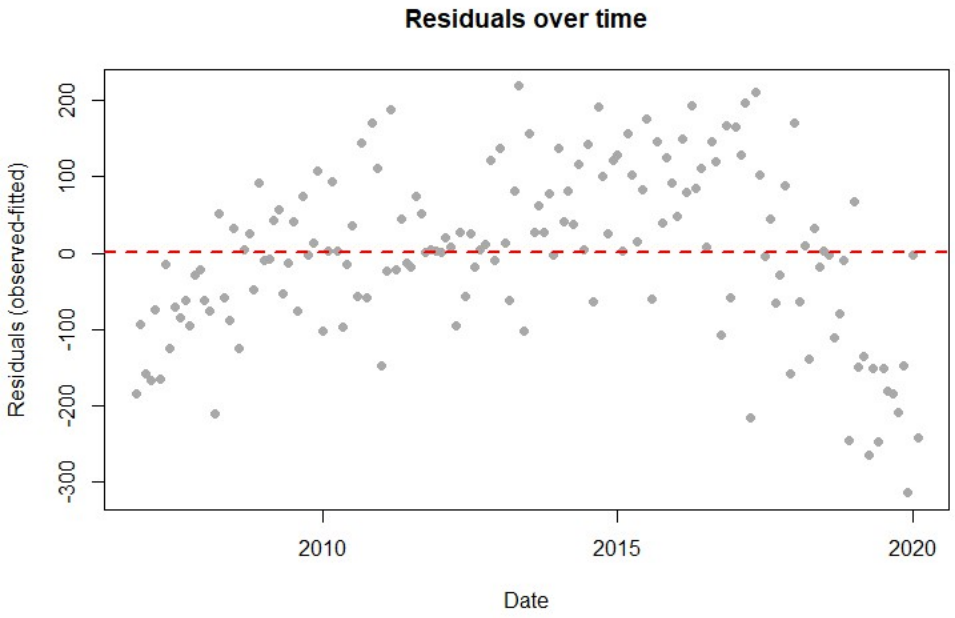


Note. The residuals over time plot is constructed from a 5 sine-cosine pair model.

Figure A3

Diagnostic Plots for Queensland Baseline Model

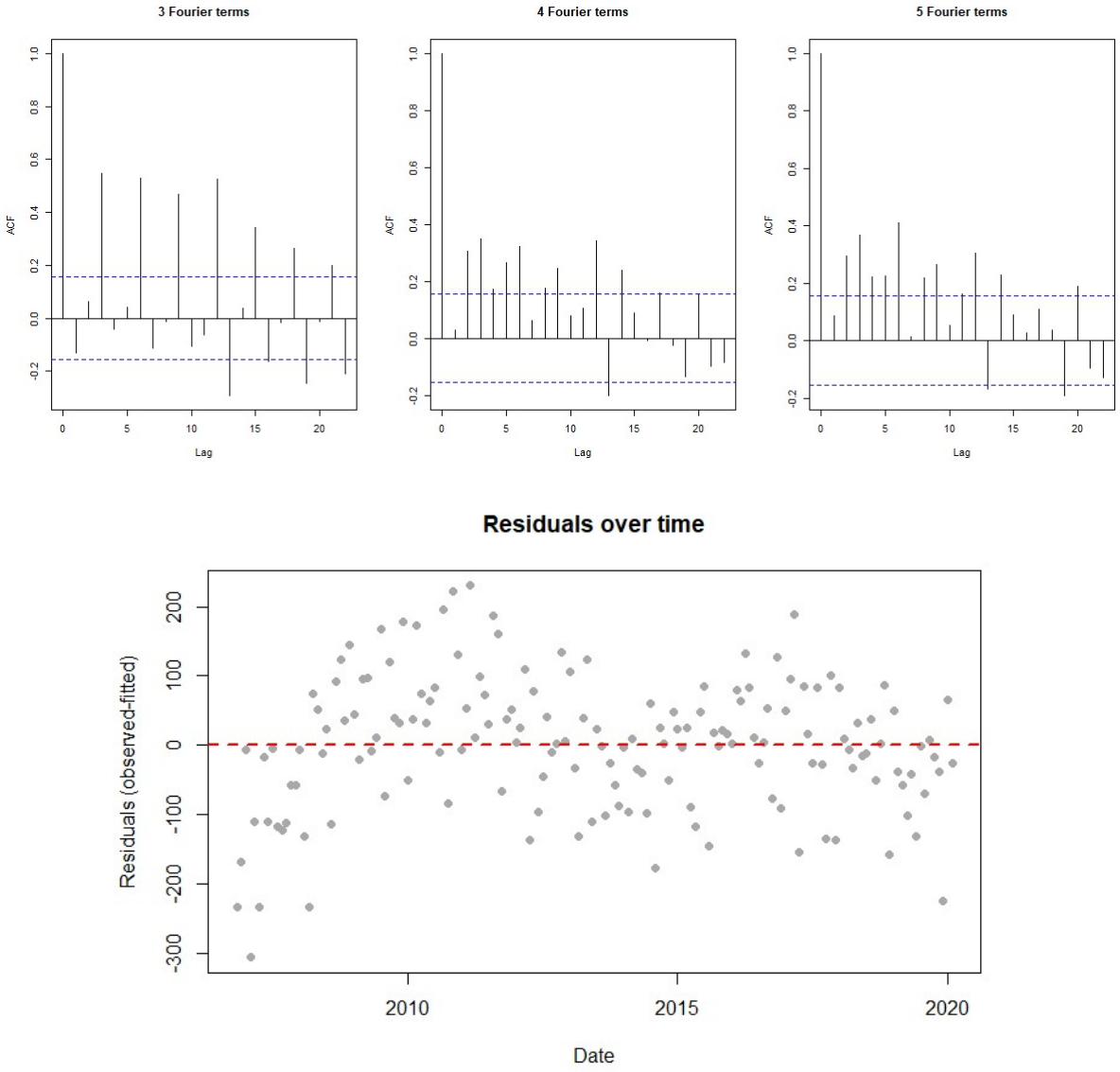




Note. The residuals over time plot is constructed from a 5 sine-cosine pair model.

Figure A4

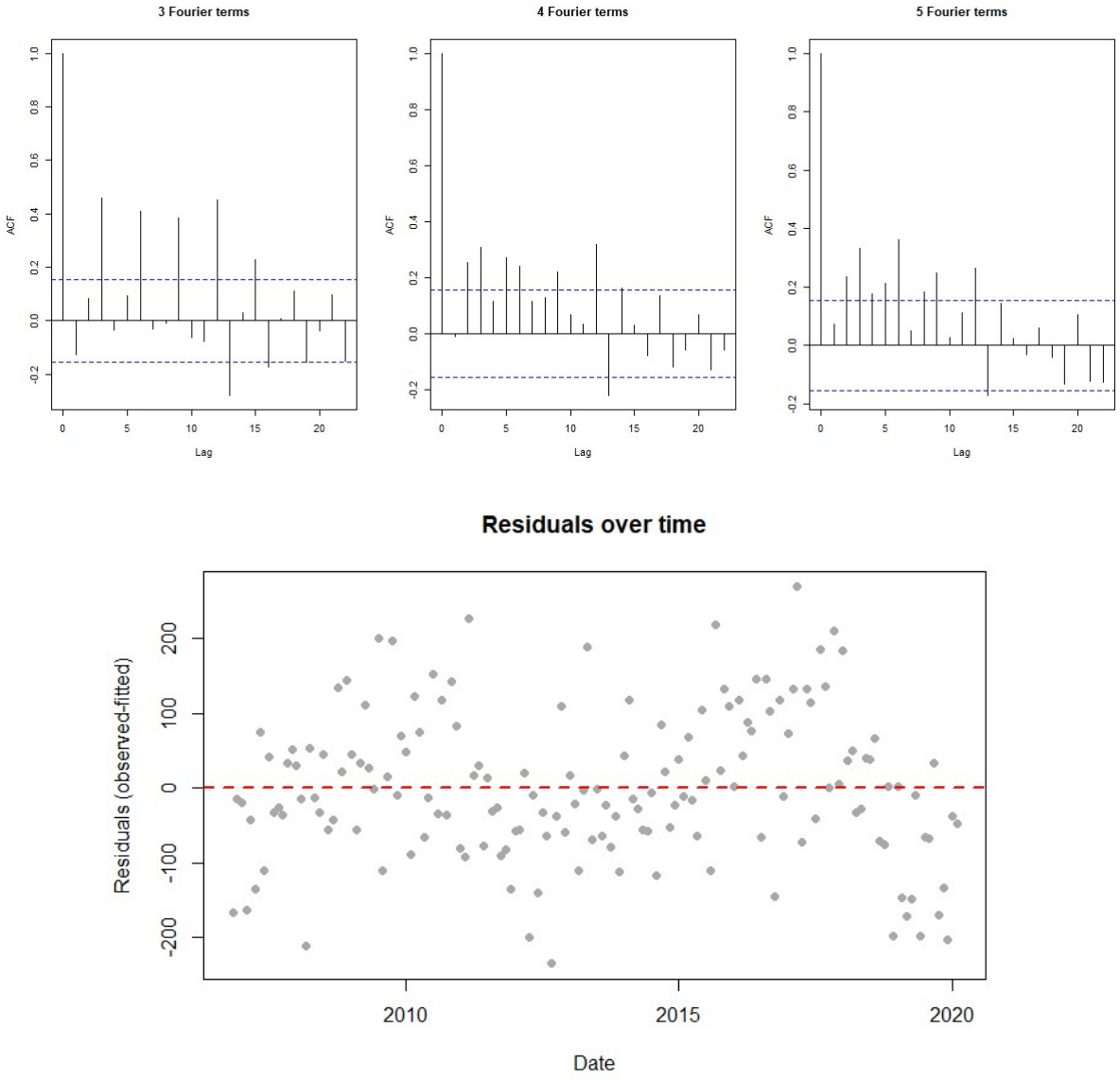
Diagnostic Plots for South Australia Baseline Model



Note. The residuals over time plot is constructed from a 5 sine-cosine pair model.

Figure A5

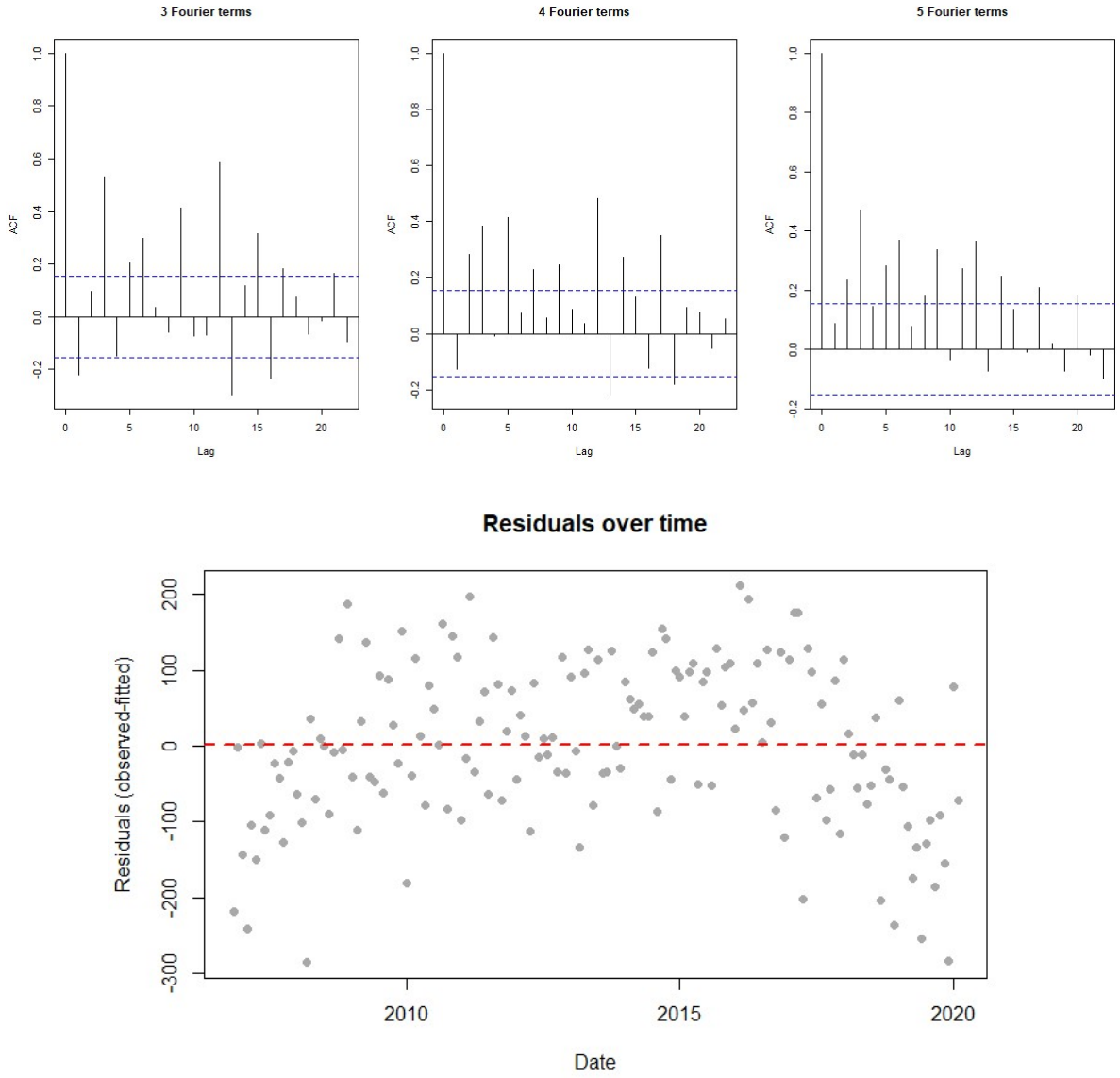
Diagnostic Plots for Tasmania Baseline Model



Note. The residuals over time plot is constructed from a 5 sine-cosine pair model.

Figure A6

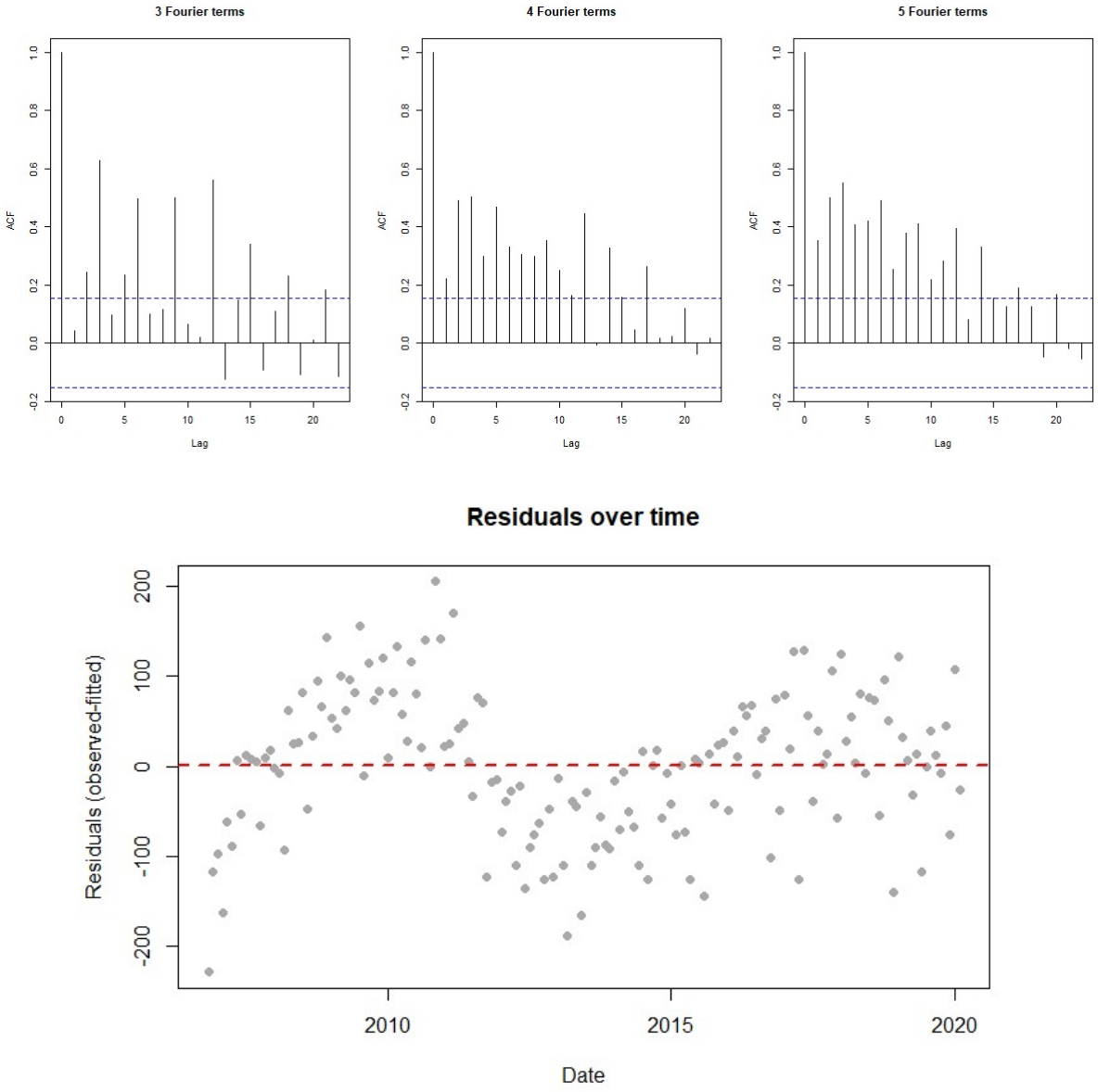
Diagnostic Plots for Victoria Baseline Model



Note. The residuals over time plot is constructed from a 5 sine-cosine pair model.

Figure A7

Diagnostic Plots for Western Australia Baseline Model



Note. The residuals over time plot is constructed from a 5 sine-cosine pair model.

Appendix B

Climate Change Related Model Estimates by State for Mar 2020 – Dec 2023

Model	Covariate		State						
			NSW	NT	QLD	SA	TAS	VIC	WA
Total MH Services (Aim 3)	Time	*Risk Ratio*	0.998	1.001	0.995	0.995	0.994	0.996	0.997
		p value	**0.008***	0.187	**5.23×10^{-8}***	**5.64×10^{-7}***	**1.70×10^{-7}***	**1.94×10^{-6}***	**1.35×10^{-4}***
	Monthly Temperature	*Risk Ratio*	0.999	0.988	1.006	1.007	0.992	0.999	0.997
		P value	0.898	0.314	0.637	0.516	0.716	0.962	0.795
	High Temperature Days	*Risk Ratio*	1.009	1.007	1.000	0.998	1.008	1.008	1.007
		p value	0.293	0.282	0.965	0.819	0.690	0.616	0.460
	Wildfire	*Risk Ratio*							0.973
		p value							0.752
Telehealth MH Services (Aim 4)	Time	*Risk Ratio*	0.997	1.009	0.991	0.990	0.983	0.977	1.002
		p value	0.411	**0.016***	**0.034***	**0.029***	**0.005***	**1.56×10^{-8}***	0.666
	Monthly Temperature	*Risk Ratio*	0.987	1.048	1.059	1.003	0.999	1.012	1.089
		P value	0.809	0.288	0.441	0.964	0.997	0.792	0.225
	High Temperature Days	*Risk Ratio*	0.991	1.012	1.008	0.933	1.098	0.931	0.933
		p value	0.870	0.578	0.802	0.289	0.442	0.266	0.365
	Wildfire	*Risk Ratio*							1.431
		p value							0.514

*$p < .05$